EPATADA 0.0.1

An Example TADA Workflow Using Penobscot Nation Data

TADA Team

2026-03-27

Source: vignettes/PenobscotNationWorkflow.Rmd

Install and Load the EPATADA R Package

First, install and load the remotes package specifying the repo. This is needed before installing EPATADA because it is only available on GitHub (not CRAN).

```
install.packages("remotes")
# Load the remotes library
library(remotes)
```

Next, install and load TADA using the remotes package. TADA R Package dependencies will also be downloaded automatically from CRAN with the TADA install. You may be prompted in the console to update dependency packages that have more recent versions available. If you see this prompt, it is recommended to update all of them (enter 1 into the console).

```
remotes::install_github("USEPA/EPATADA",
  ref = "develop",
```

On this page

Install and Load the EPATADA R Package
Retrieve Penobscot Nation data
Explore and refine results
Analyze Data Against Available Thresholds (Criteria Magnitude)
Analyze Data Against Full Criteria (Magnitude, Duration, Frequency)

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Finally, use the **library()** function to load the TADA R Package into your R session.

```
library(EPATADA)
```

Retrieve Penobscot Nation data

To start, let's find Penobscot Nation water monitoring data that was submitted to EPA's Water Quality eXchange. We are interested in Class B waters only. In this example, we already have a list of the Monitoring Location Identifiers that represent Class B waters. We can use this list of locations to query the WQP. In addition, we are interested in E. Coli data and can specify that in the characteristicName filter.

```
df_raw <- TADA_DataRetrieval(
  siteid = c(
    "PENOBSCOTINDIANNATIONDNR-130-BM1",
    "PENOBSCOTINDIANNATIONDNR-4-CH1",
    "PENOBSCOTINDIANNATIONDNR-5-DC1",
    "PENOBSCOTINDIANNATIONDNR-6-DP1",
    "PENOBSCOTINDIANNATIONDNR-76-DP3",
    "PENOBSCOTINDIANNATIONDNR-13-EM1",
    "PENOBSCOTINDIANNATIONDNR-14-EM2",
    "PENOBSCOTINDIANNATIONDNR-15-GB1",
    "PENOBSCOTINDIANNATIONDNR-16-GB2",
    "PENOBSCOTINDIANNATIONDNR-17-GB3",
    "PENOBSCOTINDIANNATIONDNR-18-GF1",
    "PENOBSCOTINDIANNATIONDNR-19-GF2",
```

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```
    "PENOBSCOTINDIANNATIONDNR-129-GWTR1",
    "PENOBSCOTINDIANNATIONDNR-20-LE1",
    "PENOBSCOTINDIANNATIONDNR-21-LE2",
    "PENOBSCOTINDIANNATIONDNR-22-LE3",
    "PENOBSCOTINDIANNATIONDNR-23-LI1",
    "PENOBSCOTINDIANNATIONDNR-24-LT1",
    "PENOBSCOTINDIANNATIONDNR-25-MD1",
    "PENOBSCOTINDIANNATIONDNR-26-MD2",
    "PENOBSCOTINDIANNATIONDNR-27-MI1",
    "PENOBSCOTINDIANNATIONDNR-28-MI2",
    "PENOBSCOTINDIANNATIONDNR-29-MI3",
    "PENOBSCOTINDIANNATIONDNR-30-MS1",
    "PENOBSCOTINDIANNATIONDNR-31-MW1",
    "PENOBSCOTINDIANNATIONDNR-32-MW2",
    "PENOBSCOTINDIANNATIONDNR-33-NL1",
    "PENOBSCOTINDIANNATIONDNR-117-OR1",
    "PENOBSCOTINDIANNATIONDNR-34-OT1",
    "PENOBSCOTINDIANNATIONDNR-35-OT2",
    "PENOBSCOTINDIANNATIONDNR-36-OT3",
    "PENOBSCOTINDIANNATIONDNR-37-PA1",
    "PENOBSCOTINDIANNATIONDNR-38-PA2",
    "PENOBSCOTINDIANNATIONDNR-39-PA3",
    "PENOBSCOTINDIANNATIONDNR-40-PA4",
    "PENOBSCOTINDIANNATIONDNR-41-PI1",
    "PENOBSCOTINDIANNATIONDNR-42-PI2",
    "PENOBSCOTINDIANNATIONDNR-43-PI3",
    "PENOBSCOTINDIANNATIONDNR-44-PS1",
    "PENOBSCOTINDIANNATIONDNR-46-SH2",
    "PENOBSCOTINDIANNATIONDNR-47-SL1",
    "PENOBSCOTINDIANNATIONDNR-81-TCOS1",
    "PENOBSCOTINDIANNATIONDNR-87-THOB",
    "PENOBSCOTINDIANNATIONDNR-135-TPIR14",
```

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```
    "PENOBSCOTINDIANNATIONDNR-103-TPIR6",
    "PENOBSCOTINDIANNATIONDNR-104-TPIR9",
    "PENOBSCOTINDIANNATIONDNR-86-TPOB",
    "PENOBSCOTINDIANNATIONDNR-91-TPUS1",
    "PENOBSCOTINDIANNATIONDNR-92-TPUS2",
    "PENOBSCOTINDIANNATIONDNR-131-VZI1",
    "PENOBSCOTINDIANNATIONDNR-132-VZTR1",
    "PENOBSCOTINDIANNATIONDNR-48-WB1",
    "PENOBSCOTINDIANNATIONDNR-49-WBU1",
    "PENOBSCOTINDIANNATIONDNR-50-WD1",
    "PENOBSCOTINDIANNATIONDNR-51-WD2",
    "PENOBSCOTINDIANNATIONDNR-75-WD3",
    "PENOBSCOTINDIANNATIONDNR-52-WE1",
    "PENOBSCOTINDIANNATIONDNR-53-WE2",
    "PENOBSCOTINDIANNATIONDNR-54-WE3",
    "PENOBSCOTINDIANNATIONDNR-56-WL1"
  ),
  characteristicName = "Escherichia coli",
  maxrecs = 350000,
  ask = FALSE,
  applyautoclean = TRUE
)
```

```
## Checking what data is available. This may take a moment.
```

```
## Downloading WQP query results. This may take some time dependi
```

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```
## "PENOBSCOTINDIANNATIONDNR-117-OR1", "PENOBSCOTINDIANNATIONDNR-1
## "PENOBSCOTINDIANNATIONDNR-87-THOB", "PENOBSCOTINDIANNATIONDNR-1
## "PENOBSCOTINDIANNATIONDNR-50-WD1", "PENOBSCOTINDIANNATIONDNR-51
```

```
## Data successfully downloaded. Running TADA_AutoClean function.
```

```
## TADA_Autoclean: creating TADA-specific columns.
```

```
## TADA_Autoclean: handling special characters and coverting TADA
```

```
## TADA_Autoclean: converting TADA.LatitudeMeasure and TADA.Longit
```

```
## TADA_Autoclean: harmonizing synonymous unit names (m and meters
```

```
## TADA_Autoclean: updating deprecated (i.e. retired) characterist
```

```
## No deprecated characteristic names found in dataset.
```

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```
## TADA_Autoclean: creating TADA.ComparableDataIdentifier field fo
```

```
## NOTE: This version of the TADA package is designed to work with
```

Explore and refine results

Review unique monitoring projects:

```
unique(df_raw$ProjectName)
```

```
## [1] "Baseline Water Quality Monitoring"   "Penobscot River Rest
```

Generate pie chart:

```
EPATADA::TADA_FieldValuesPie(df_raw, field = "MonitoringLocationTy
```

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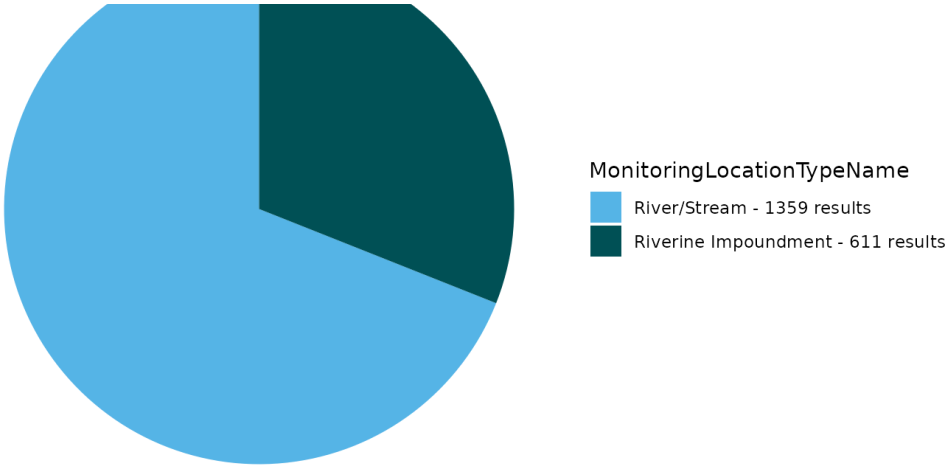
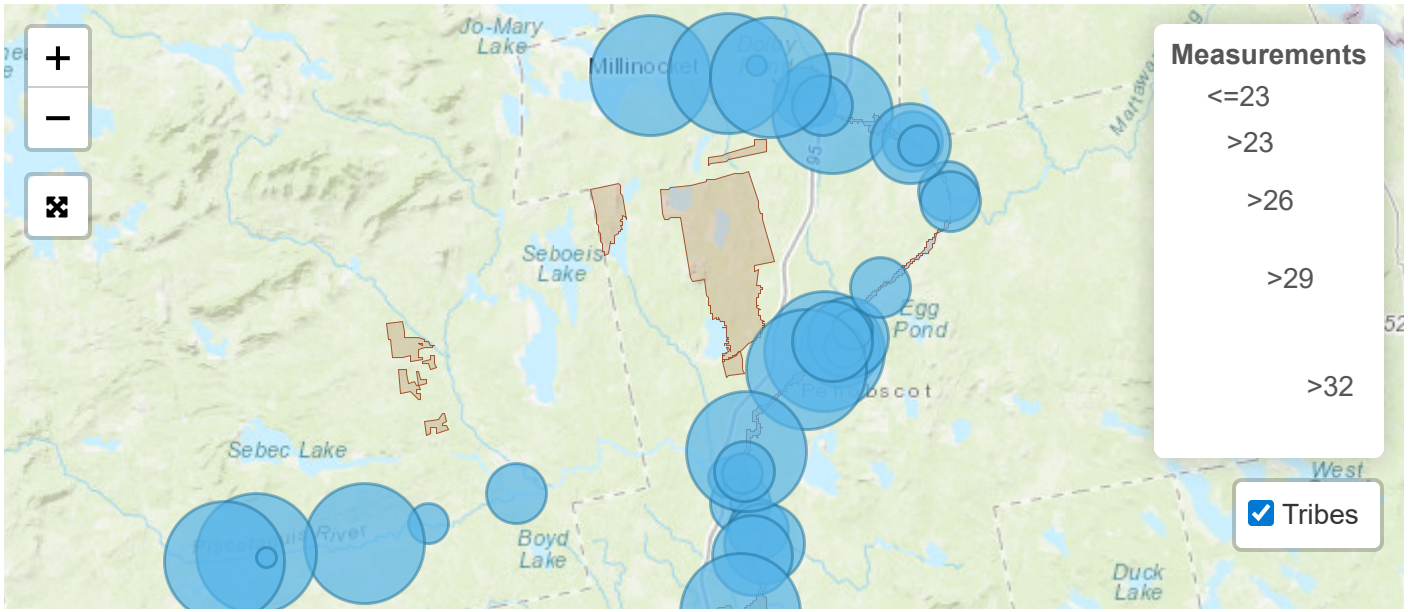


Use `TADA_OverviewMap` to generate a map:

```
EPATADA::TADA_OverviewMap(df_raw)
```



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Review and remove duplicate results if present:

```
df_flag <- EPATADA::TADA_FindPotentialDuplicatesSingleOrg(df_raw)
```

```
## TADA_FindPotentialDuplicatesSingleOrg: 13 groups of potentially
```

Select a small number of columns to review duplicates:

```
TADA_TableExport(subset(
  df_flag,
  TADA.SingleOrgDupGroupID != "Not a duplicate",
  select = c(ResultIdentifier,
             ActivityStartDate,
             TADA.ActivityMediaName,
             TADA.MonitoringLocationIdentifier,
             TADA.CharacteristicName,
             TADA.ResultMeasureValue,
             TADA.ResultMeasure.MeasureUnitCode,
             ResultAnalyticalMethod.MethodName)
))
```

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DA.ResultMeasureValue	TADA.ResultMeasure.MeasureUnitCode
8	CFU/100ML
8	CFU/100ML
1203	CFU/100ML
1203	CFU/100ML
20	CFU/100ML

Showing 1 to 5 of 26 entries

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Remove duplicates:

```
df_clean <- dplyr::filter(df_flag, TADA.SingleOrgDup.Flag == "Uni
```

Run key TADA quality control flagging functions. Review carefully and consider removing flagged results.

```
df_flag <- EPATADA::TADA_RunKeyFlagFunctions(
  df_clean,
```

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```
## All characteristic/fraction combinations are valid in your data
```

```
## TADA_FlagSpeciation: Rows with Suspect speciations have been fl
```

```
## TADA_FindQCActivities: Quality control samples have been remove
```

```
## TADA_FlagMeasureQualifierCode: Dataframe does not include any :
```

Flag results above and below thresholds. Review carefully and consider removing.

```
df_flag <- EPATADA::TADA_FlagAboveThreshold(df_flag,
  clean = FALSE,
  flaggedonly = FALSE
)
```

```
## TADA_FlagAboveThreshold: Returning the dataframe with flags. Co
```

```
df_flag <- EPATADA::TADA_FlagBelowThreshold(df_flag,
  clean = FALSE,
```

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```
## TADA_FlagBelowThreshold: No data below the WQX Lower Threshold
```

For now, let's assume we reviewed the flagged results and want to move forward with all of them:

```
df_clean <- df_flag
```

Harmonize synonyms if found:

```
df_clean <- EPATADA::TADA_HarmonizeSynonyms(df_clean)
```

```
## Warning: Your dataframe contains suspect metadata combinations
```

```
##                  Flag_Column Result Count
## 1 TADA.MethodSpeciation.Flag         1957
```

Generate table:

```
EPATADA::TADA_FieldValuesTable(df_clean, field = "ActivityTypeCod
```

```
##            Value Count
```

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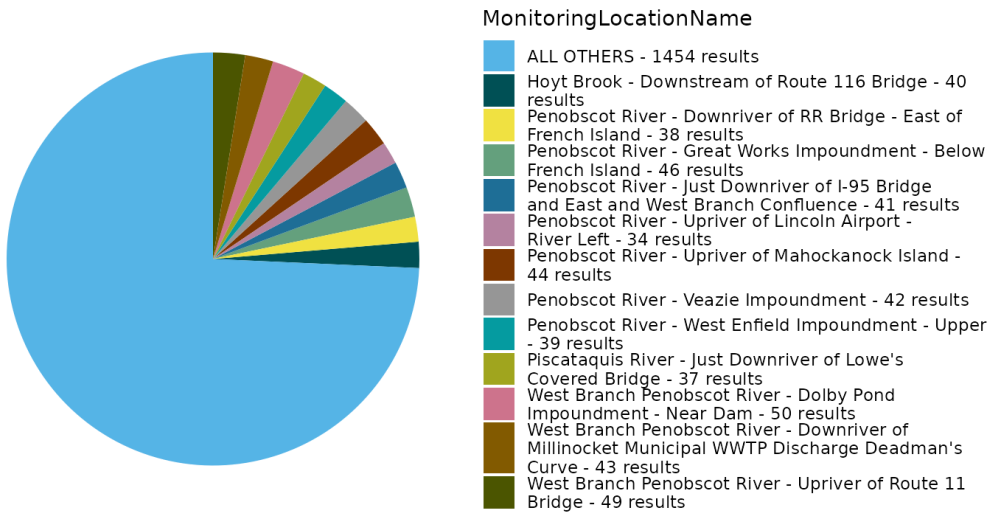
Generate pie chart:

```
EPATADA::TADA_FieldValuesPie(df_clean, field = "MonitoringLocatio
```



Remove non-numeric results:

```
df_clean <- EPATADA::TADA_ConvertSpecialChars(
  df_clean,
  col = "TADA.ResultMeasureValue",
  clean = TRUE
)
```

Review the number of sites and records for each characteristic:

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Search:

	TADA.CharacteristicName	n_sites	n_records
1	ESCHERICHIA COLI	65	1957

Showing 1 to 1 of 1 entries Previous 1 Next

Generate statistics for each Monitoring Location:

```
EPATADA::TADA_TableExport(EPATADA::TADA_Stats(df_clean,
                                              group_cols = c("TADA.Comparal
                                                             "TADA.Monitor:
```

```
## TADA_IDCensoredData: No censored data detected in your datafrar
```

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Search:

	TADA.ComparableDataIdentifier	TADA.MonitoringLocationIder
1	ESCHERICHIA COLI_NONE_NONE_CFU/100ML	PENOBSCOTINDIANNATIONDNR-100-TP
2	ESCHERICHIA COLI_NONE_NONE_CFU/100ML	PENOBSCOTINDIANNATIONDNR-103-TP

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4	ESCHERICHIA COLI_NONE_NONE_CFU/100ML	PENOBSCOTINDIANNATIONDNR-103-TP
5	ESCHERICHIA COLI_NONE_NONE_CFU/100ML	PENOBSCOTINDIANNATIONDNR-113-GW

Showing 1 to 5 of 65 entries

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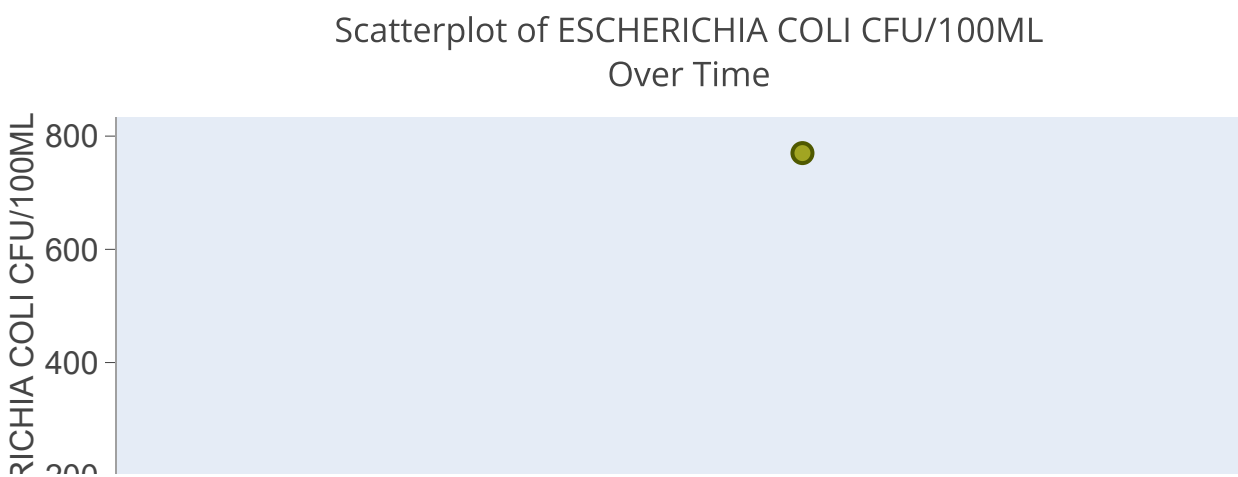
Generate scatter plot for E. coli (top four monitoring location by number of results will be displayed):

```
EPATADA::TADA_GroupedScatterplot(df_clean,
                                 group_col = "MonitoringLocationNa
```

```
## TADA_GroupedScatterplot: No 'groups' selected for MonitoringLoc
```



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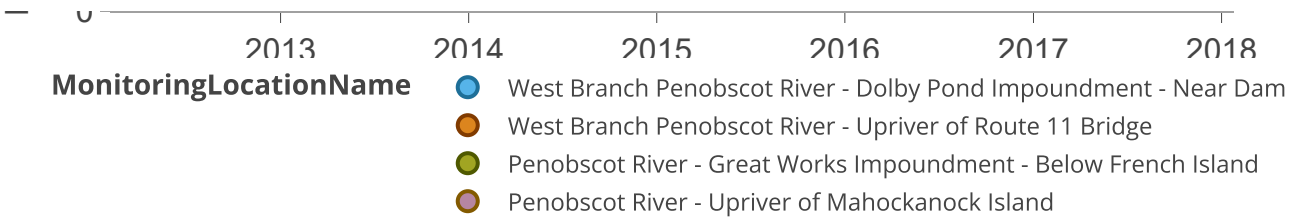
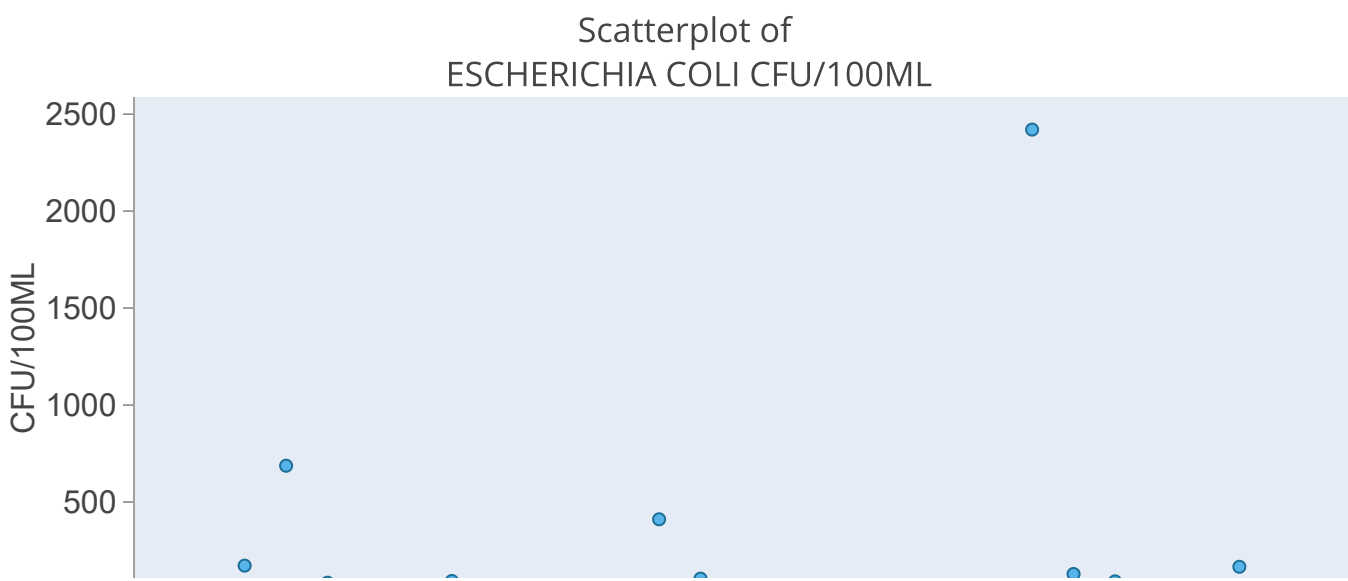


Filter to a single site and continue exploring E. coli:

```
df_singleML <- dplyr::filter(
  df_clean,
  TADA.MonitoringLocationIdentifier %in% c(
    "PENOBSCOTINDIANNATIONDNR-123-TPIR16"
  )
)

EPATADA::TADA_Scatterplot(df_singleML)
```



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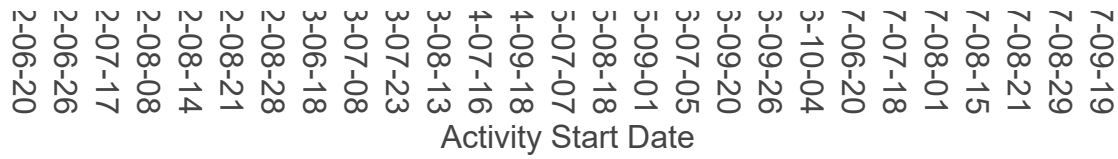

Analyze Data Against Available Thresholds (Criteria Magnitude)

Let's check if any results are above the EPA 304A recommended maximum criteria magnitude (see: [2012 Recreational Water Quality Criteria Fact Sheet](#)).

CRITERIA ELEMENTS	**Recommendation 1** Estimated Illness Rate 36/1,000		**Recommendation 2** Estimated Illness Rate 32/1,000	
Indicator	GM (cfu/100 mL)	STV (cfu/100 mL)	GM (cfu/100 mL)	STV (cfu/100 mL)
Enterococci (marine & fresh)	35	130	30	110
E. coli (fresh)	126	410	100	320

If interested, you can find other state, tribal, and EPA 304A criteria in [EPA's Criteria Search Tool](#).

Let's check if any individual results exceed 320 CFU/100mL (the magnitude component of the EPA recommendation 2 criteria for ESCHERICHIA COLI).

Add column with comparison to criteria mag (excursions):

```
df_singleML <- df_singleML |>
  dplyr::mutate(meets_criteria_mag = ifelse(TADA.ResultMeasureValu
```

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```
## [1] "Yes" "No"
```

Review a table of the results:

```
# review subset
df_clean_subset_review <- df_singleML |>
  dplyr::select(
    MonitoringLocationIdentifier, MonitoringLocationName,
    OrganizationFormalName, ActivityStartDate, TADA.ResultMeasure\
    TADA.ResultMeasure.MeasureUnitCode, meets_criteria_mag
  )

EPATADA::TADA_TableExport(df_clean_subset_review)
```

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Search:

	MonitoringLocationIdentifier	MonitoringLocationName	O
1	PENOBSCOTINDIANNATIONDNR-123-TPIR16	Piscataquis River - Downriver of Route 23 Bridge	Pe
2	PENOBSCOTINDIANNATIONDNR-123-TPIR16	Piscataquis River - Downriver of Route 23 Bridge	Pe
3	PENOBSCOTINDIANNATIONDNR-123-TPIR16	Piscataquis River - Downriver of Route 23 Bridge	Pe

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5	PENOBSCOTINDIANNATIONDNR-123-TPIR16	Piscataquis River - Downriver of Route 23 Bridge	Pe

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Generate stats table. Review percentiles. Less than 5% of results fall above 23.4 CFU/100mL and over 98% of results fall below 1310 CFU/100m

```
EPATADA::TADA_TableExport(EPATADA::TADA_Stats(df_singleML))
```

```
## TADA_IDCensoredData: No censored data detected in your datafrar
```

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Search:

	TADA.ComparableDataIdentifier	Location_Count	Measurem
1	ESCHERICHIA COLI_NONE_NONE_CFU/100ML	1	

Showing 1 to 1 of 1 entries Previous 1 Next

Generate a scatterplot. Three result values are above the threshold.

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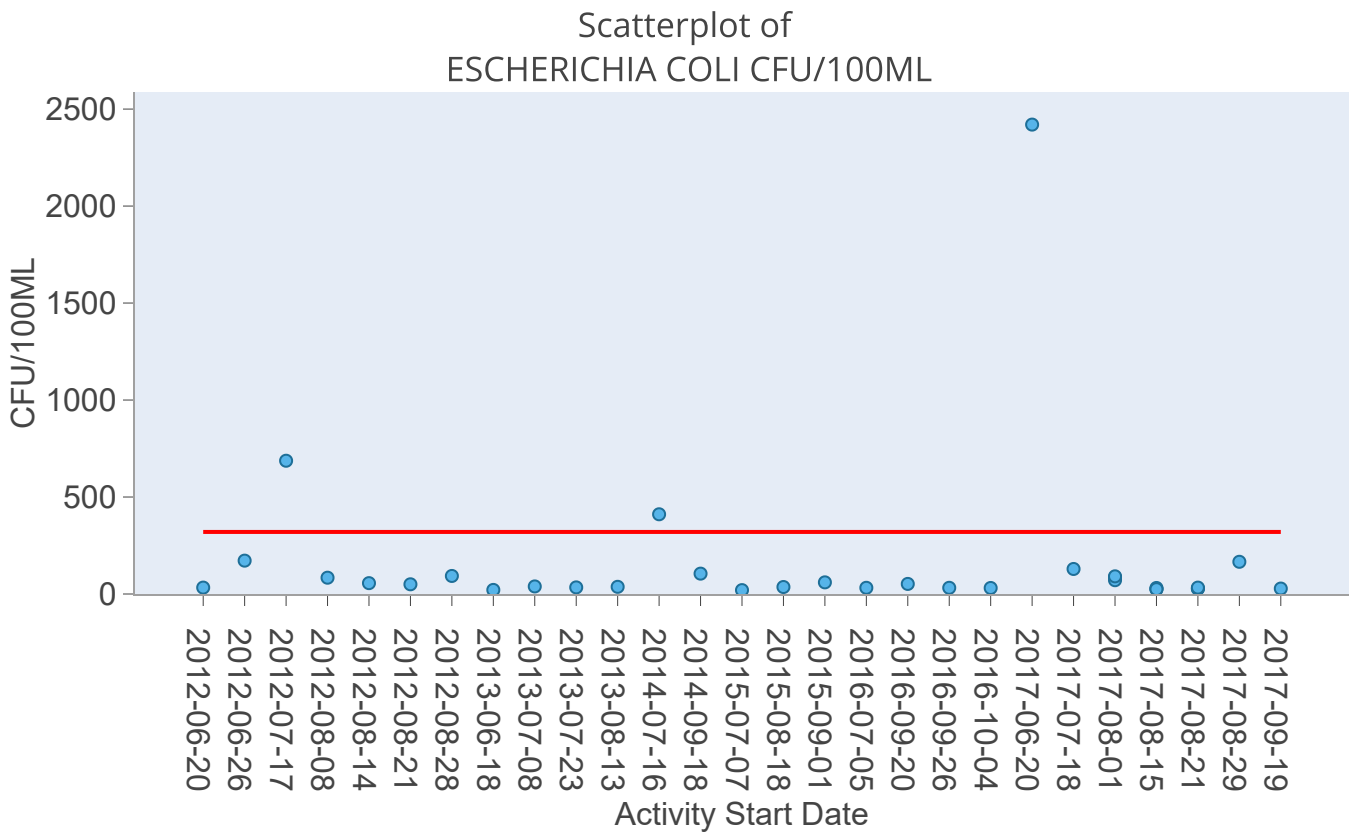
```
    y = 320,
    x = c(min(df_singleML$ActivityStartDate), max(df_singleML$Act
    inherit = FALSE,
    showlegend = FALSE,
    line = list(color = "red"),
    hoverinfo = "none"
  )
```



Generate a histogram.

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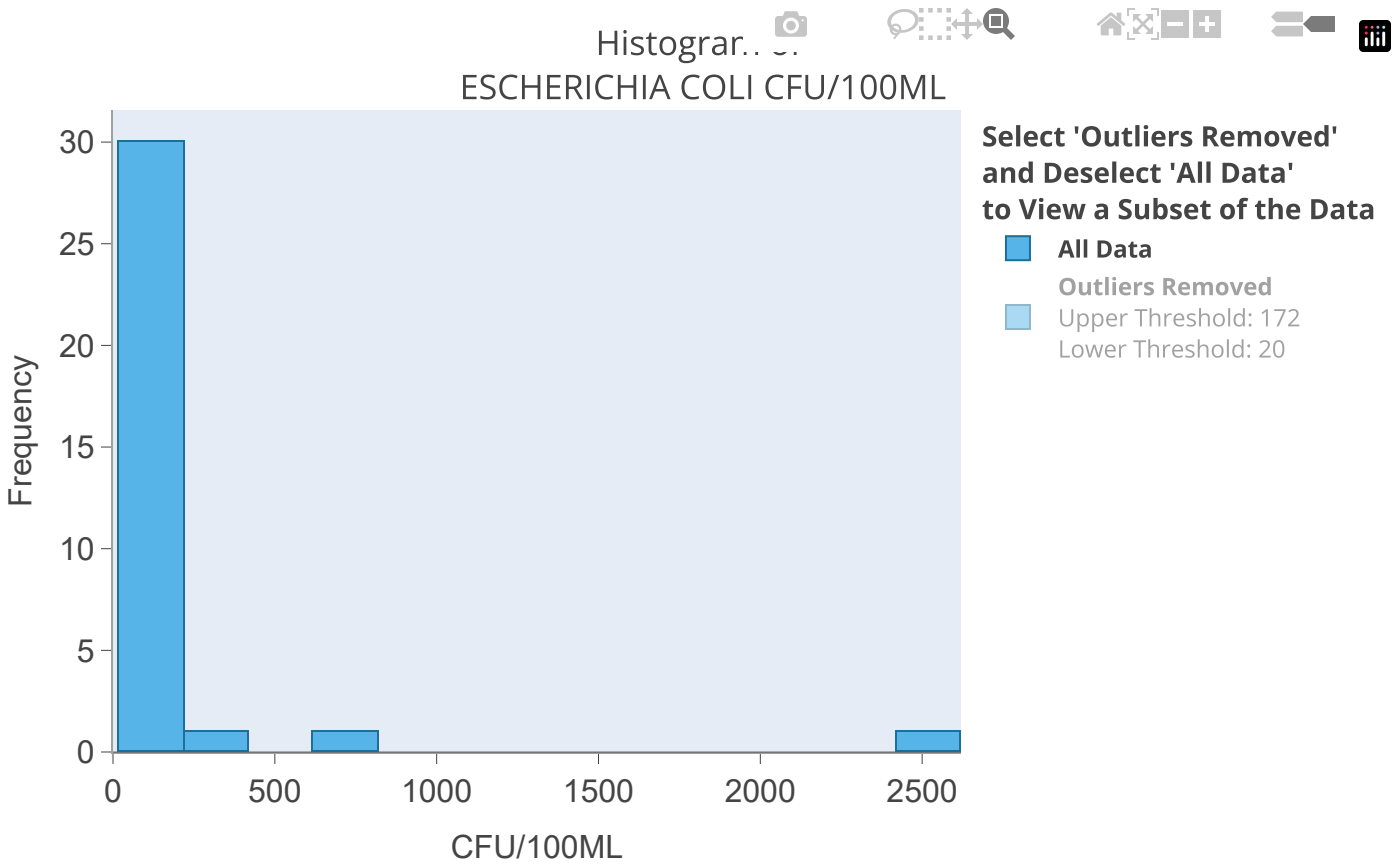

`TADA_Boxplot` can be useful for identifying skewness and percentiles.

```
EPATADA::TADA_Boxplot(df_singleML, id_cols = "TADA.ComparableData
```

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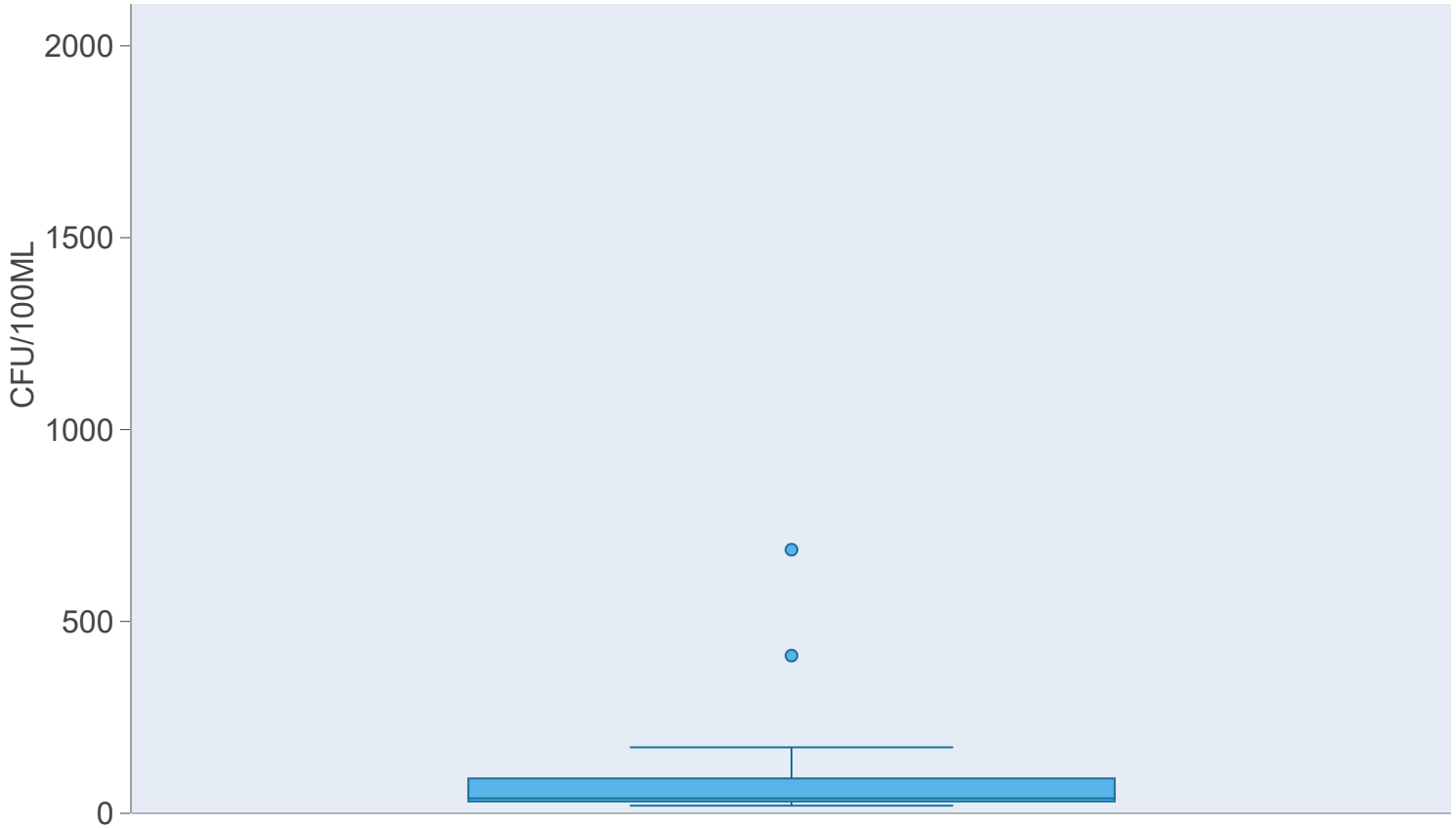

Analyze Data Against Full Criteria (Magnitude, Duration, Frequency)

Now, let's look at **Maine's Water Quality Standards for Class B** surface waters. Two different criteria can be assessed using E. Coli data:

The magnitude components of these criteria are:

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Recreation

This information is available in EPA's Criteria Search Tool. Additional details specific to Maine are available.

We can pull this information into R using the TADA_DefineCriteriaMethodology() function:

```
ME_criteria_ecoli = TADA_DefineCriteriaMethodology(df_singleML, o
```

```
## TADA_DefineCriteriaMethodology: auto_assign = TRUE was selecte
```

```
## TADA_DefineCriteriaMethodology: auto_assign = TRUE was selecte
```

```
## TADA_DefineCriteriaMethodology: auto_assign = TRUE was selecte
```

```
## TADA_UsesForAnalysis: auto_assign == TRUE was selected, assign
```

```
##
```

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```
## TADA_DefineCriteriaMethodology: displayUniqueId == FALSE was se
```

Filter output to include only Class B waters:

```
ME_criteria_ecoli_classB = unique(subset(ME_criteria_ecoli,
       CST.Use == "FRESH SURFACE WATERS - CLASS B"))

TADA_TableExport(ME_criteria_ecoli_classB)
```

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Search:

	ATTAINS.OrganizationIdentifier	TADA.CharacteristicName	A
1	MEDEP	ESCHERICHIA COLI	E
2	MEDEP	ESCHERICHIA COLI	E
3	MEDEP	ESCHERICHIA COLI	E
4	MEDEP	ESCHERICHIA COLI	E

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In this table, both criteria have been applied by MEDEP to assess PRIMARY CONTACT RECREATION and SECONDARY CONTACT RECREATION designated uses in the past.

When reviewing the table, you'll notice that additional detail is needed to perform a complete water quality assessment. This information is available on pg. 3 of the WQS documentation (links to these documents are included in ME_criteria_ecoli_classB). This detailed methodology can be added manually to the TADA Criteria and Methods template (e.g., ME_criteria_ecoli_classB).

For simplicity, let's focus on finding and added a few critical components: the DurationValue, DurationUnit, DurationMethod, FreqValue, FreqMethod, SeasonStartDate, and SeasonEndDate.

- DurationValue: The numeric value component of the length of time in which a waterbody can be exposed to a magnitude of a parameter without negatively impacting its designated use.
- DurationUnit: The units component of the length of time in which a waterbody can be exposed to a magnitude of a parameter without negatively impacting its designated use.
- DurationMethod: The specific aggregation calculation of samples that are collected during a duration period. FreqValue Optional User Supplied Criteria The numeric value of how often a

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- FreqMethod: How often a magnitude value can be exceeded percentage or number of times a magnitude value can be exceeded over a specified duration period.
- SeasonStartDate: The start date of the season in which assessments are done for during a calendar year.
- SeasonEndDate: The end date of the season in which assessments are done for during a calendar year.

```
ME_criteria_ecoli_classB_final <- ME_criteria_ecoli_classB |>
  dplyr::mutate(
    DurationValue   = 90,
    DurationUnit    = "n-day",
    DurationMethod = dplyr::if_else(dplyr::row_number() <= 2, "ge
    FreqValue       = dplyr::if_else(dplyr::row_number() <= 2, NA_
    FreqMethod      = dplyr::if_else(dplyr::row_number() <= 2, NA_
    SeasonStartDate = "4/15/2025",
    SeasonEndDate   = "10/31/2025"
  )

TADA_TableExport(ME_criteria_ecoli_classB_final)
```

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Search:

ATTAINS.OrganizationIdentifier	TADA.CharacteristicName	A

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2	MEDEP	ESCHERICHIA COLI	E
3	MEDEP	ESCHERICHIA COLI	E
4	MEDEP	ESCHERICHIA COLI	E

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You may notice the rows repeat the two criteria. This is because the same two criteria are used to assess primary and secondary contact recreation. For Class B waters (both primary and secondary recreational uses), the Escherichia coli (E. coli) criteria are as follows:

Between April 15th and October 31st, the number of E. coli bacteria in Class B waters may not exceed:

1. A geometric mean of 64 CFU or MPN per 100 milliliters over a 90-day interval.
2. 236 CFU or MPN per 100 milliliters in more than 10% of the samples in any 90-day interval.

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these criteria are not met, then a waterbody is considered impaired.

The EPATADA R package does not yet contain the functionality to assess data against WQS. We are currently working on that functionality - in the future, we will be able to analyze data using the information provided in a Criteria and Methods Template (e.g., ME_criteria_ecoli_classB_final). We are also working on an R Shiny application that will take this Criteria and Methods Template as an input and perform the analysis. Stay tuned for these new features!

In the meantime, here is an example workflow to perform the analysis in this example. This analysis will run if a minimum of 3 samples are available in any 90-day period for each year.

```
# ----------------------------------------------------------------
# Class B assessment
# - TADA.CharacteristicName == "ESCHERICHIA COLI"
# - Date: ActivityStartDate
# - Units: uses TADA.ResultMeasure.MeasureUnitCode (expects MPN/1
# - Windows: 90-day rolling, within Apr 15–Oct 31
# ----------------------------------------------------------------
assess_ecoli_classB <- function(
  data,
  gm_limit           = 64,   # geometric mean limit (per 100 mL)
  single_limit       = 236,  # threshold for percentile check
  season_start_month = 4,
  season_start_day   = 15,
  season_end_month   = 10,
  season_end_day     = 31,
  window_days        = 90,
  min_n_geomean      = 3,    # minimum samples for criterion A
```

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```
) {
  df <- data |>
    dplyr::filter(`TADA.CharacteristicName` == "ESCHERICHIA COLI"
    dplyr::transmute(
      site       = as.character(`TADA.MonitoringLocationIdentifie
      site_name  = `TADA.MonitoringLocationName`,
      date       = as.Date(ActivityStartDate),  # strictly Activi
      value      = suppressWarnings(as.numeric(`TADA.ResultMeasur
    ) |>
    dplyr::filter(!is.na(site), !is.na(date), !is.na(value)) |>
    dplyr::arrange(site, date)

  if (nrow(df) == 0) {
    stop("No rows available after filtering to ESCHERICHIA COLI w
  }

  season_bounds <- function(y) {
    list(
      start = lubridate::make_date(y, season_start_month, season_
      end   = lubridate::make_date(y, season_end_month, season_en
    )
  }

  windows <- df |>
    dplyr::mutate(season_year = lubridate::year(date)) |>
    dplyr::group_by(site, site_name, season_year) |>
    dplyr::group_modify(~{
      y  <- .y$season_year[[1]]
      sb <- season_bounds(y)
      wd <- as.integer(window_days)

      # All window end dates fully inside the season
```

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```
# Season-filtered data for this group
season_dat <- .x |>
  dplyr::filter(date >= sb$start, date <= sb$end) |>
  dplyr::arrange(date)

# If no samples in-season, return skeleton
if (nrow(season_dat) == 0) {
  return(tibble::tibble(
    window_start = w_end_seq - lubridate::days(wd - 1),
    window_end   = w_end_seq,
    n            = 0L,
    geomean      = NA_real_,
    p90          = NA_real_,
    pass_A       = NA,
    pass_B       = NA
  ))
}

dts  <- season_dat$date
vals <- season_dat$value
logs <- log(pmax(vals, zero_offset))

m       <- length(w_end_seq)
n_vec   <- integer(m)
gm_vec  <- rep(NA_real_, m)
p90_vec <- rep(NA_real_, m)
pass_A  <- rep(NA, m)
pass_B  <- rep(NA, m)

# Two-pointer over sample rows [L..R]
L <- 1L
R <- 0L
```

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```
  w_end   <- w_end_seq[i]
  w_start <- w_end - lubridate::days(wd - 1)

  while (R < n_samples && dts[R + 1L] <= w_end) R <- R + 1L
  while (L <= R && dts[L] < w_start) L <- L + 1L

  if (L <= R) {
    idx_range <- L:R
    nwin <- length(idx_range)
    n_vec[i] <- nwin

    # Criterion A: geometric mean
    if (nwin >= min_n_geomean) {
      gm <- exp(mean(logs[idx_range]))
      gm_vec[i] <- gm
      pass_A[i] <- gm <= gm_limit
    } else {
      gm_vec[i] <- NA_real_
      pass_A[i] <- NA
    }

    # Criterion B: 90th percentile (type = quantile_type)
    if (nwin >= min_n_exceed) {
      p90 <- as.numeric(stats::quantile(vals[idx_range], pr
      p90_vec[i] <- p90
      pass_B[i] <- p90 <= single_limit
    } else {
      p90_vec[i] <- NA_real_
      pass_B[i] <- NA
    }
  } else {
    n_vec[i]  <- 0L
```

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```
        pass_B[i] <- NA
      }
    }

    tibble::tibble(
      window_start = w_end_seq - lubridate::days(wd - 1),
      window_end   = w_end_seq,
      n            = n_vec,
      geomean      = gm_vec,
      p90          = p90_vec,
      pass_A       = pass_A,
      pass_B       = pass_B
    )
  }) |>
  dplyr::ungroup()

summary <- windows |>
  dplyr::group_by(site, site_name, season_year) |>
  dplyr::summarise(
    n_windows    = dplyr::n(),
    n_eval_A     = sum(!is.na(pass_A)),
    n_eval_B     = sum(!is.na(pass_B)),
    any_fail_A   = any(pass_A == FALSE, na.rm = TRUE),
    any_fail_B   = any(pass_B == FALSE, na.rm = TRUE),
    attain_A     = ifelse(n_eval_A > 0, !any_fail_A, NA),
    attain_B     = ifelse(n_eval_B > 0, !any_fail_B, NA),
    overall_bool = dplyr::case_when(
      is.na(attain_A) & is.na(attain_B) ~ NA,
      is.na(attain_A)                   ~ attain_B,
      is.na(attain_B)                   ~ attain_A,
      TRUE                              ~ (attain_A & attain_B)
    ),
```

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```
      ),
      worst_geomean = if (all(is.na(geomean))) NA_real_ else max(
      worst_p90     = if (all(is.na(p90)))     NA_real_ else max(
      .groups = "drop"
    ) |>
    dplyr::select(-overall_bool)

  list(windows = windows, summary = summary)
}

# Example:
res <- assess_ecoli_classB(df_singleML)
TADA_TableExport(head(res$windows))
```

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Search:

	site	site_name	season_year	window
1	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2012	2012-04-1
2	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2012	2012-04-1

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	123-TPIR16	ROUTE 23 BRIDGE		
4	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2012	2012-04-18
5	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2012	2012-04-19

Showing 1 to 5 of 6 entries Previous 1 2 Next

```
TADA_TableExport(res$summary)
```

Copy CSV Excel PDF Search:

	site	site_name	season_year	n_windc
1	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2012	
2	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF	2013	

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3	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2014
4	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2015
5	PENOBSCOTINDIANNATIONDNR-123-TPIR16	PISCATAQUIS RIVER - DOWNRIVER OF ROUTE 23 BRIDGE	2016

Showing 1 to 5 of 6 entries Previous 1 2 Next

Disclaimer: This example workflow is not meant to be representative of Penobscot Nation's or Maine's assessment process.